

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: **EEE101**

Course: **BASIC ELECTRICAL ENGINEERING**

QP No. :U005R-1

Duration: 3 hours

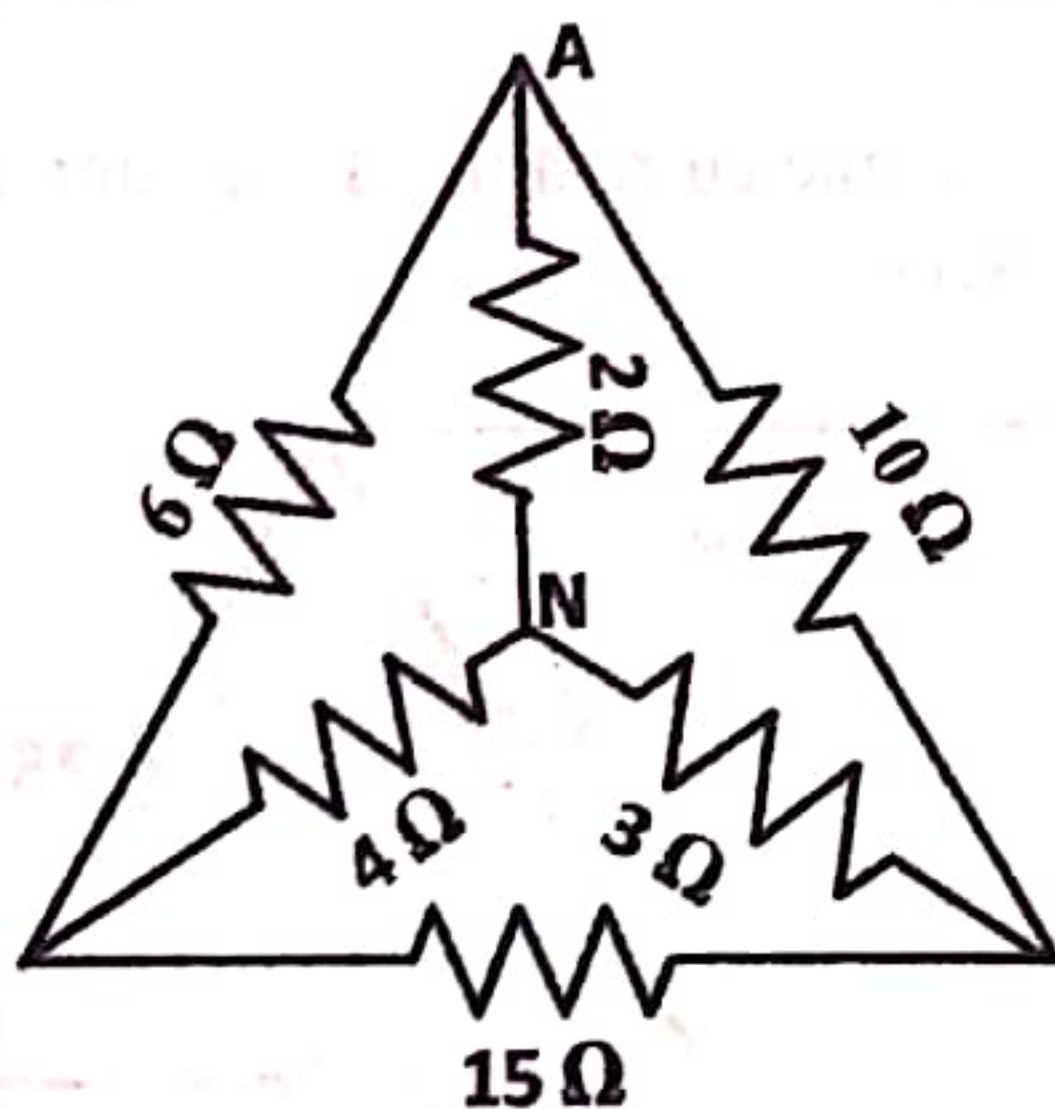
Max. Marks:100

PART – A

Answer all the questions

$10 \times 2 = 20$ Marks

1. Calculate the equivalent resistance between A and N



2. An alternating voltage has the equation $v = 169.7 \sin 377t$. What are the values of r.m.s voltage and the instantaneous voltage when $t=3$ ms?
3. Calculate the resistance and the inductance or capacitance in series for each of the following impedances (i) $10 + j15 \Omega$ and (ii) $50 \angle 30^\circ$. Assume the frequency to be 60 Hz.
4. Draw the phasor diagram of the RLC circuit assuming $X_L > X_C$.

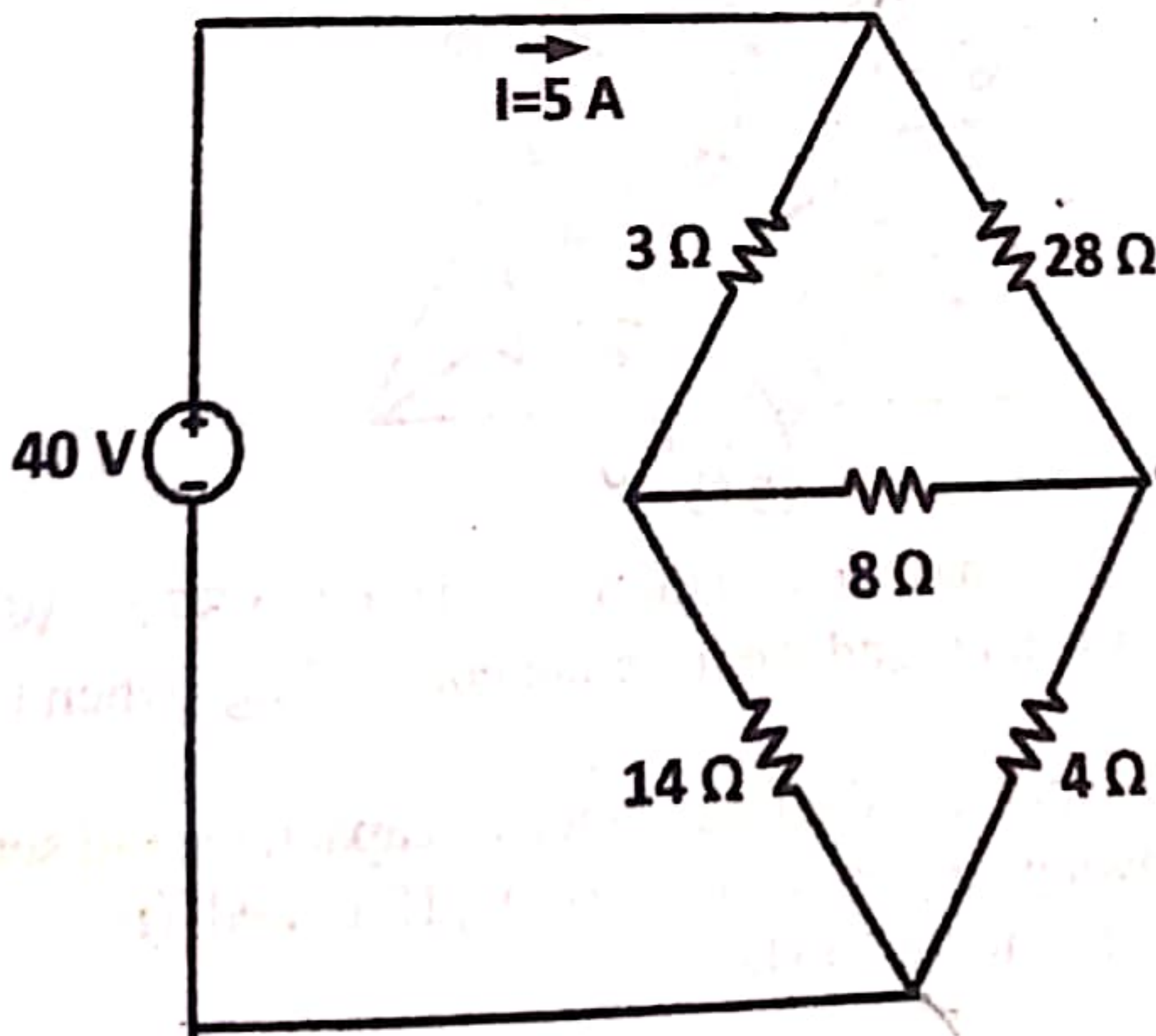
5. Draw the circuit model of the separately excited D.C. generator.
6. Why are starters required for D.C. motors?
7. State the two laws for deducing the direction of the induced e.m.f.
8. Define the voltage regulation of the transformer.
9. Draw the circuit diagram of staircase wiring.
10. Three coils, each having a resistance of $20\ \Omega$ and an inductive reactance of $15\ \Omega$, are connected in star to a $400\ \text{V}$, 3-phase, $50\ \text{Hz}$ supply. Calculate the line current.

PART - B

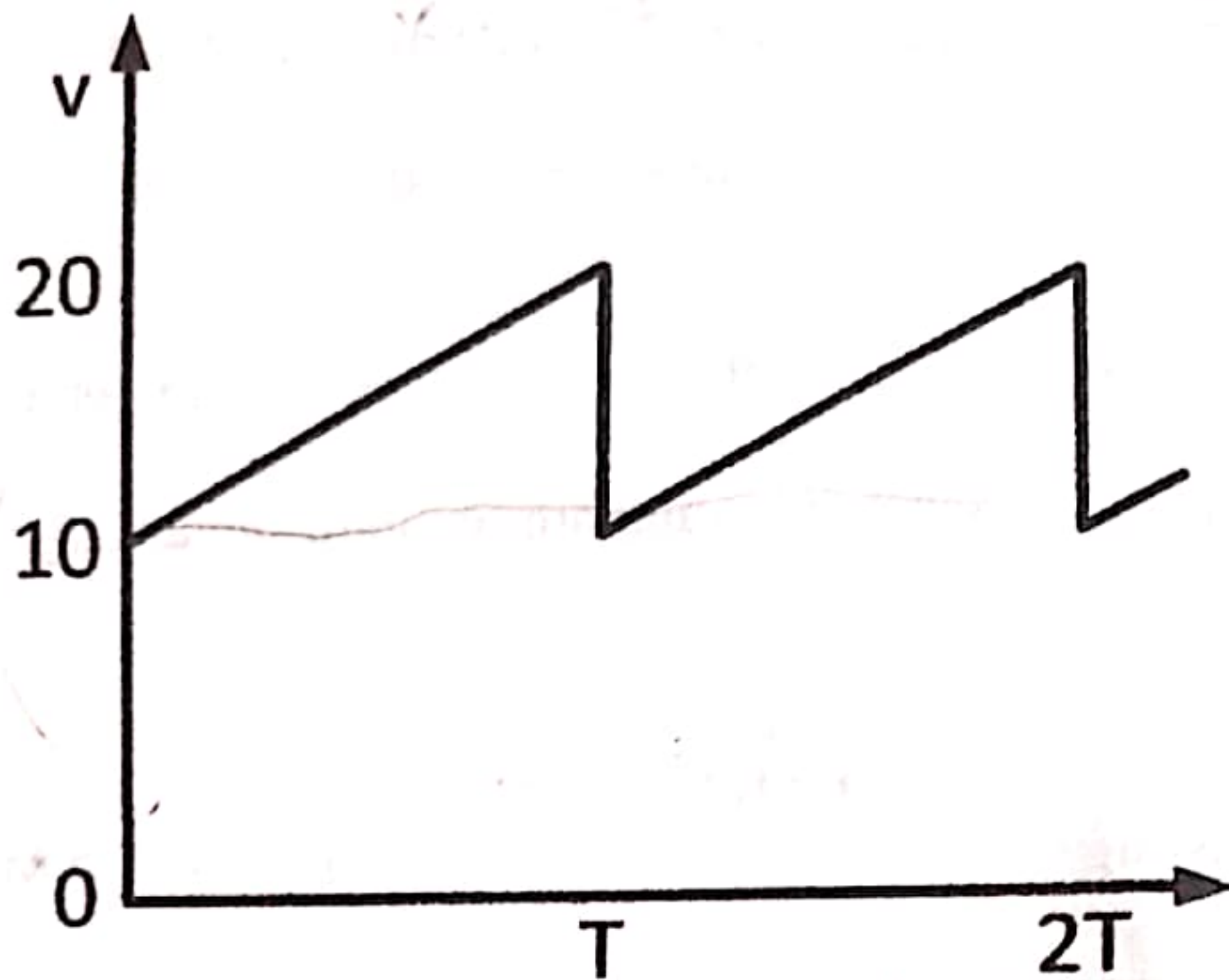
Answer any four questions

$4 \times 15 = 60$ Marks

11. Using mesh analysis, find the current in $8\ \Omega$ resistor and verify the same using nodal analysis.



12. a. Find the form factor and peak factor of the waveform shown in the figure (10)



- b. Compare electric and magnetic circuits. (5)

13. a. i. A steel ring having a mean circumference of 750 mm and a cross-sectional area of 500 mm^2 , is wound with a magnetizing coil of 120 turns. Using the following data, calculate the current required to set up a magnetic flux of $630 \mu\text{Wb}$ in the ring.

Flux density (T)	0.9	1.1	1.2	1.26
Magnetic field strength (AT/m)	260	450	600	720

- ii. The air gap in a magnetic circuit is 1.1 mm long and 2000 mm^2 in cross-section. Calculate the reluctance of the air gap and the m.m.f to send a flux of $700 \mu\text{Wb}$ across the air gap.

(8)

- b. Draw the general arrangement of a four-pole d.c. Machine and label the parts. (7)

14. Derive the e.m.f. Equation of transformer and discuss the losses and efficiency of the transformer. (15)
15. a. Consider two coils, A and B, with self-inductance L_1 and L_2 . The mutual inductance between the coils is M . Derive an expression for the coefficient of coupling between coils in terms of M , L_1 and L_2 . (10)
- b. What are the general safety requirements under the Indian Electricity Rules, 1956? (5)
16. a. Draw the structure of the electric supply system from generation to load. (5)
- b. With the help of a schematic diagram, explain the arrangement of pipe earthing. (10)

PART – C

Answer the following

1 × 20 = 20 Marks

17. a. A domestic consumer has two loads with impedance $Z_1 = 28.57 + j13.84 \Omega$ and $Z_2 = 38.22 + j23.69 \Omega$. They are connected in parallel with a 230 V, 50 Hz single-phase supply. (14)
- i. Calculate the power drawn by both loads.
 - ii. Compute the consumer's monthly energy consumption if both loads operate for 5 hours daily.
 - iii. What electricity bill will the consumer have if the unit cost of electricity is ₹6.0?
- b. A conductor, 500 mm long, is moved at a uniform speed at right angles and to a uniform magnetic field having a density of 0.4 T. if the e.m.f. Generated in the conductor is 2 V, and the conductor forms part of a closed circuit having a resistance of 0.5Ω , calculate
- i. the velocity of the conductor in m/s .
 - ii. the force acting on the conductor in Newtons.
 - iii. the work done in Joules when the conductor has moved 600 mm. (6)